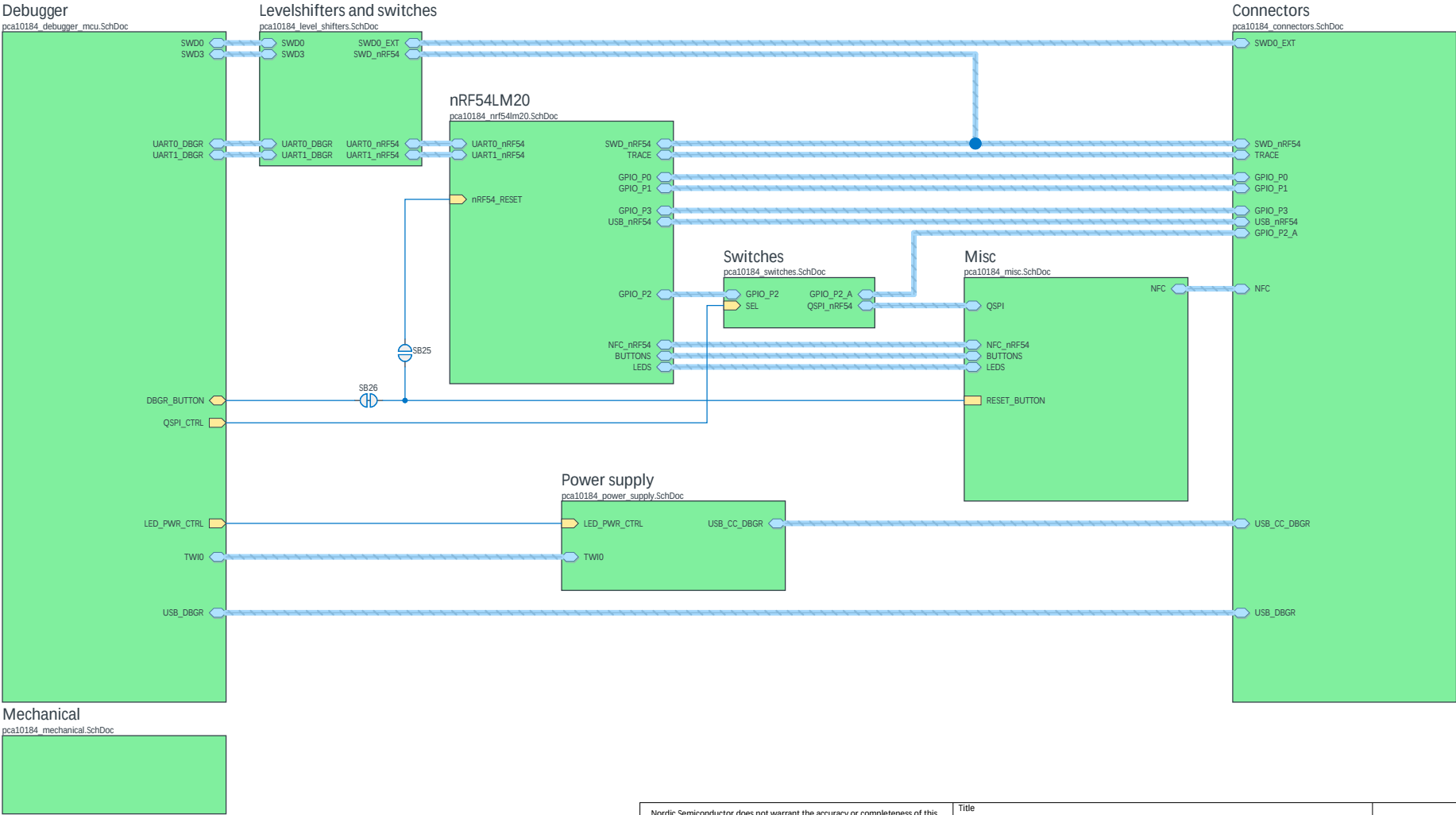


Nordic Semiconductor ASA

nRF54LM20-DK (PCA10184)



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Title
nRF54LM20 Development Kit

Size
A3

Date: 10.02.2026
File: pca10184_top_level.SchDoc
Classification: PUBLIC

Revision
0.7.0

Sheet 1 of 9
Drawn By: GETR



A

B

C

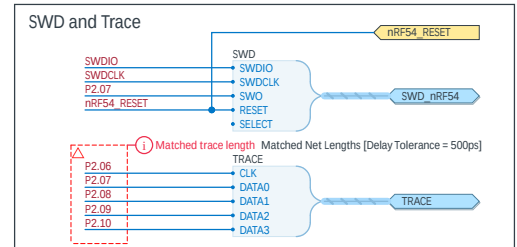
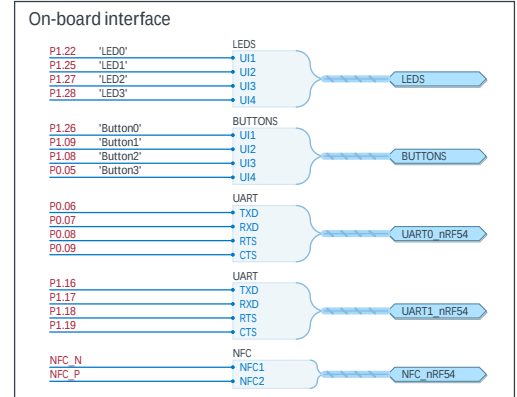
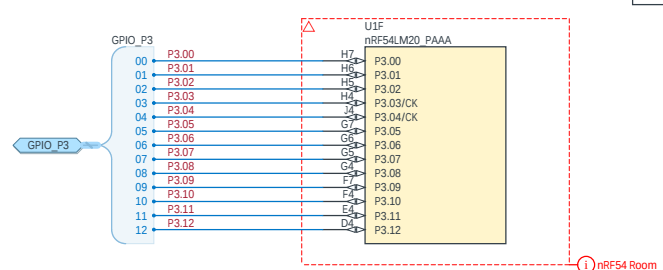
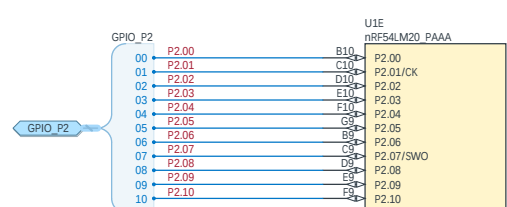
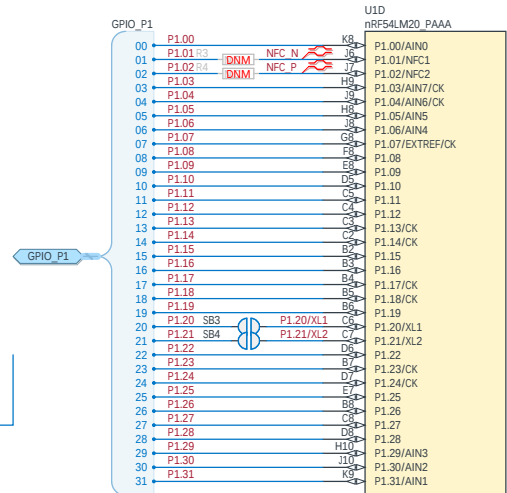
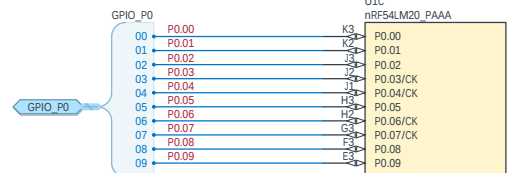
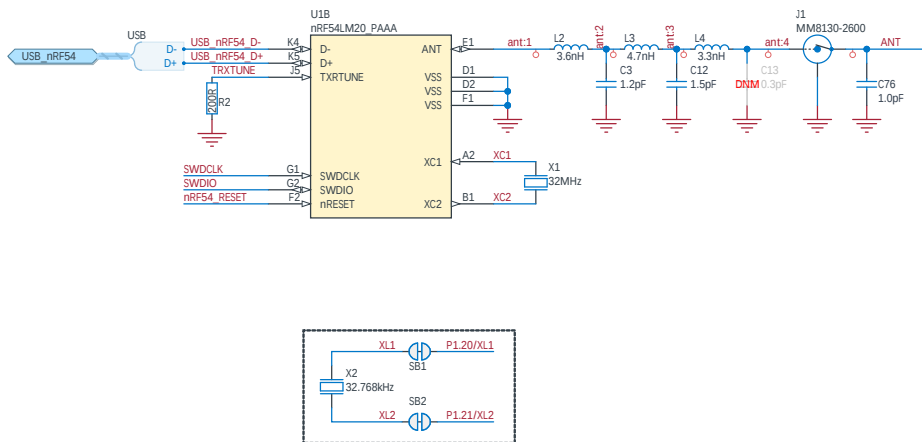
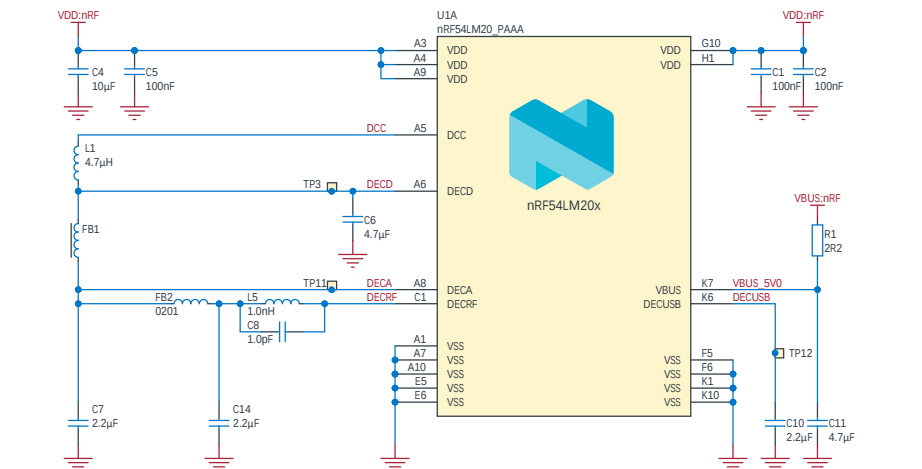
D

A

B

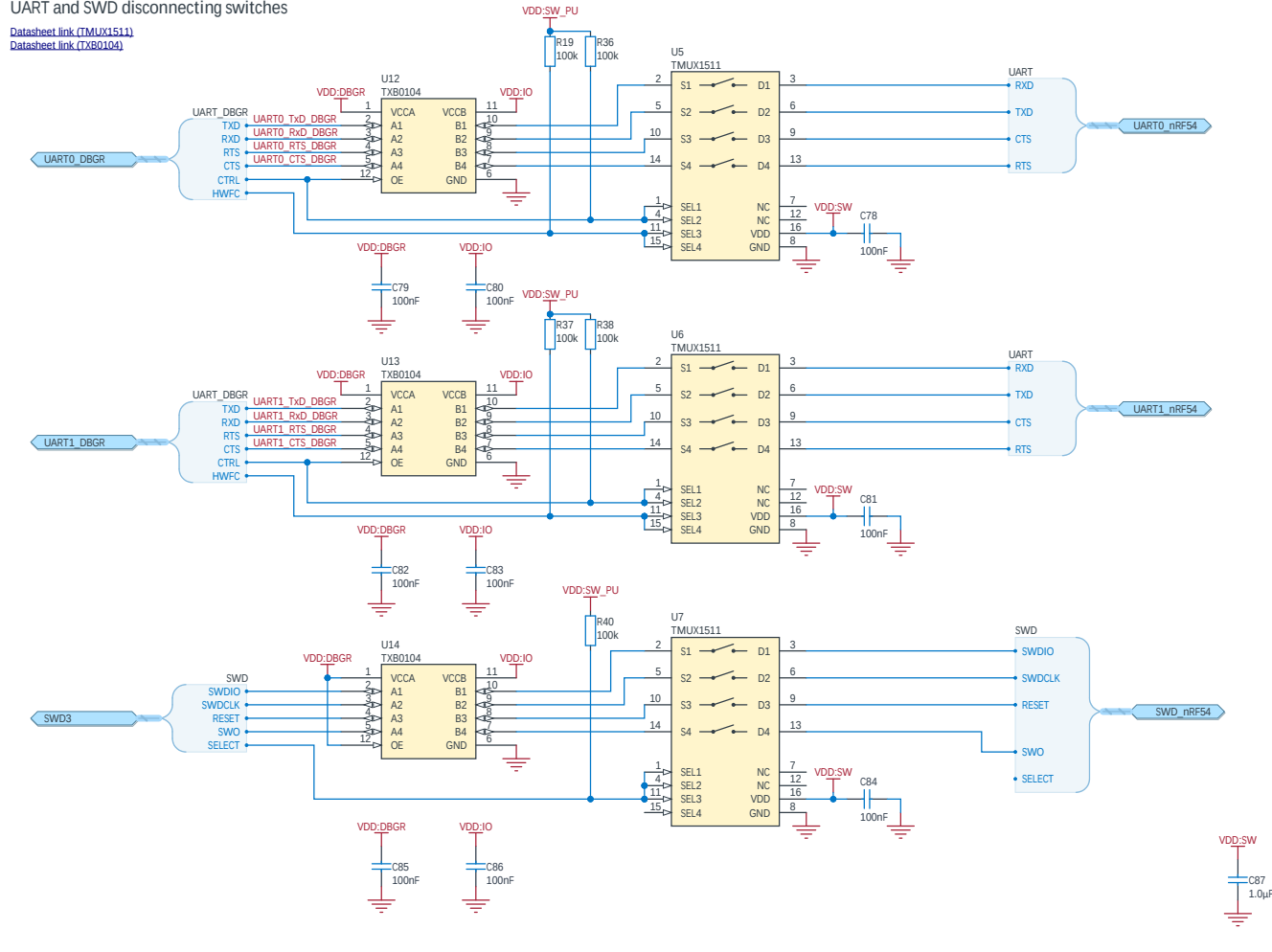
C

D

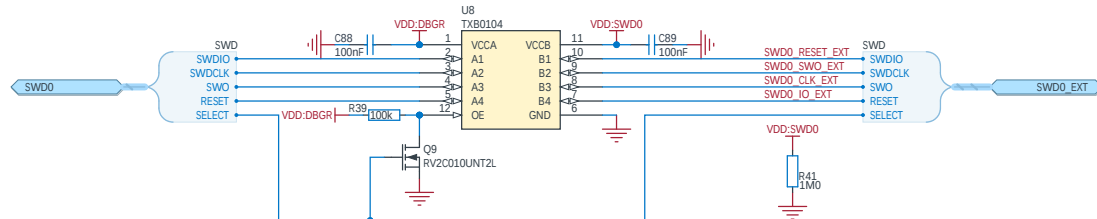


UART and SWD disconnecting switches

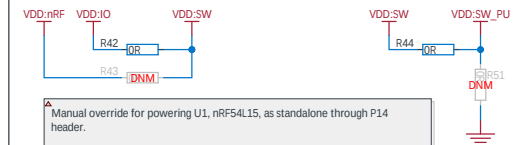
Datasheet link (TMUX1511)
Datasheet link (TXB0104)



SWD Debug Out level shifter



Analog switches manual override



Manual override for powering U1, nRF54L15, as standalone through P14 header.

R42, R44 mounted: Default configuration. DK requires power from USB connector.

R43, R51 mounted: Analog switches are powered from P14. Analog switches U5, U6, U7, U15 and U16 disconnects all signals. No external power needed from USB connector

Title
nRF54LM20 Development Kit - Level Shifters

Size
A3

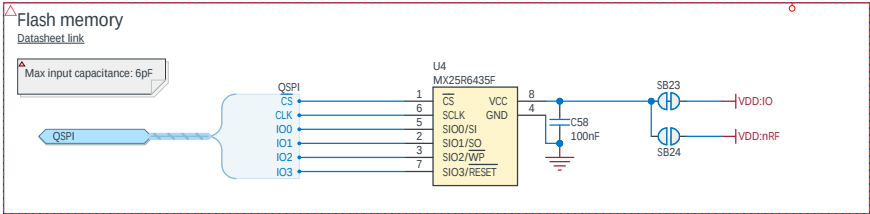
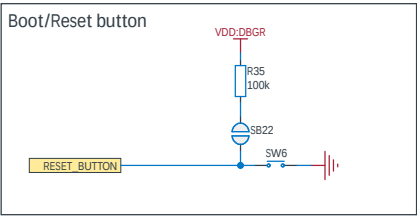
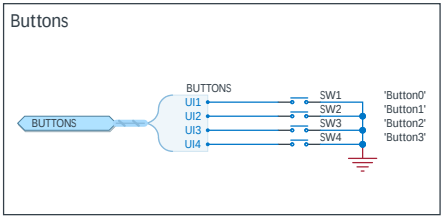
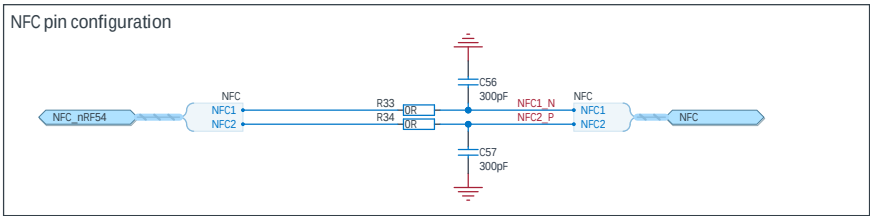
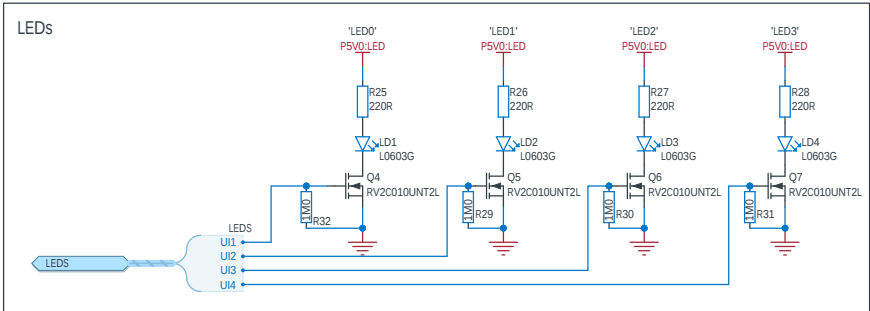
PCB Assembly Number
PCA10184

Revision
0.7.0

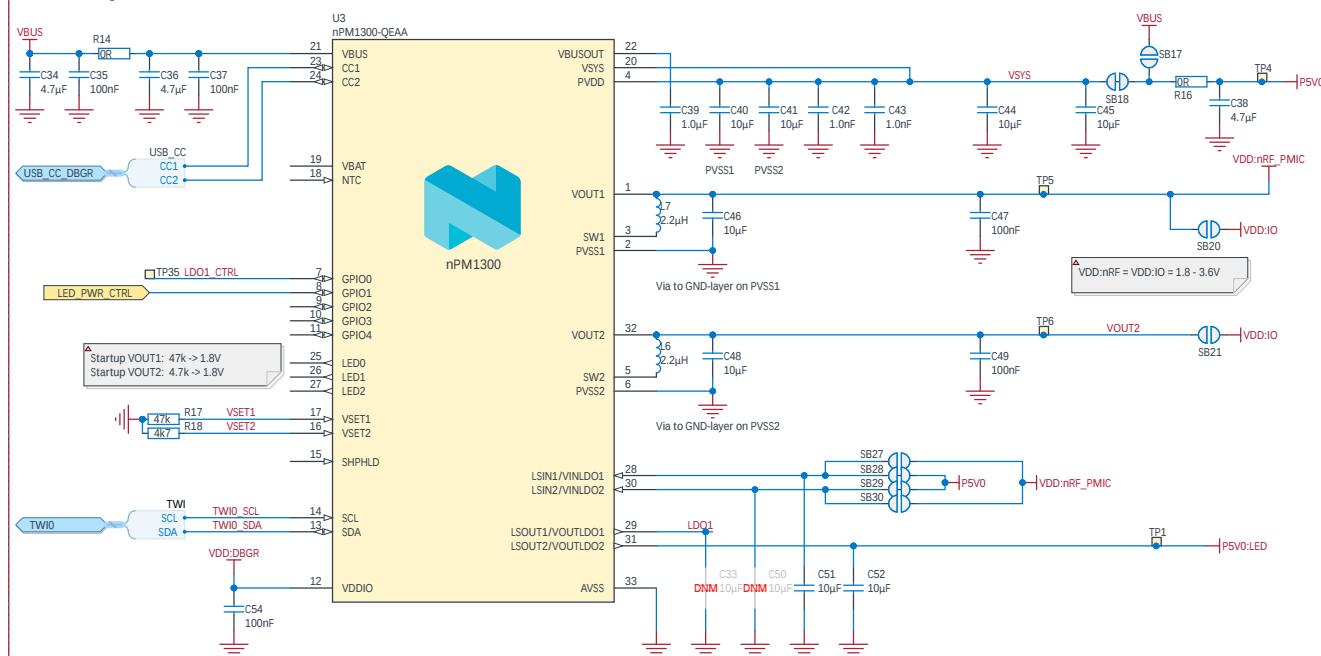
Date: 10.02.2026
File: pca10184_level_shifters.SchDoc
Classification: PUBLIC

Sheet 4 of 9
Drawn By: GETR

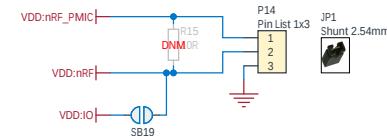




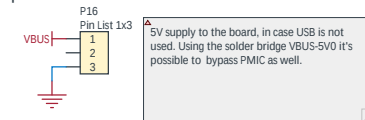
Power management



Current Measurement



External power source

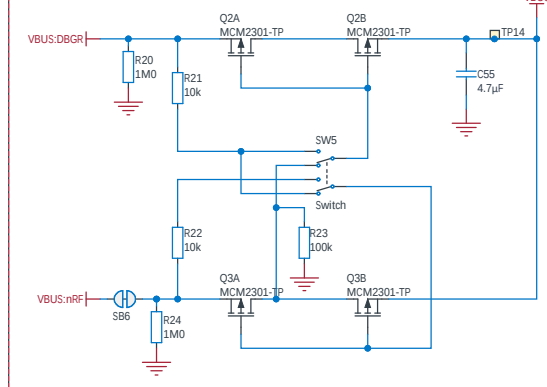


nRF54LM20 settings:

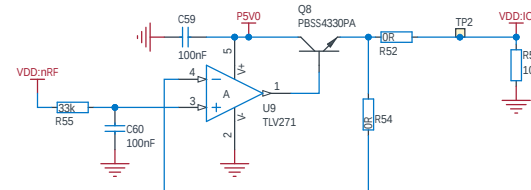
- VOUT1: 1.8V - 3.3V, max 200mA
- VOUT2: 1.8V - 3.3V, max 200mA
- LS2: Switch for LEDs power. GPIO1 control

VSS1 current limit: 900mA

Power Switch



VDD:nRF supply buffer



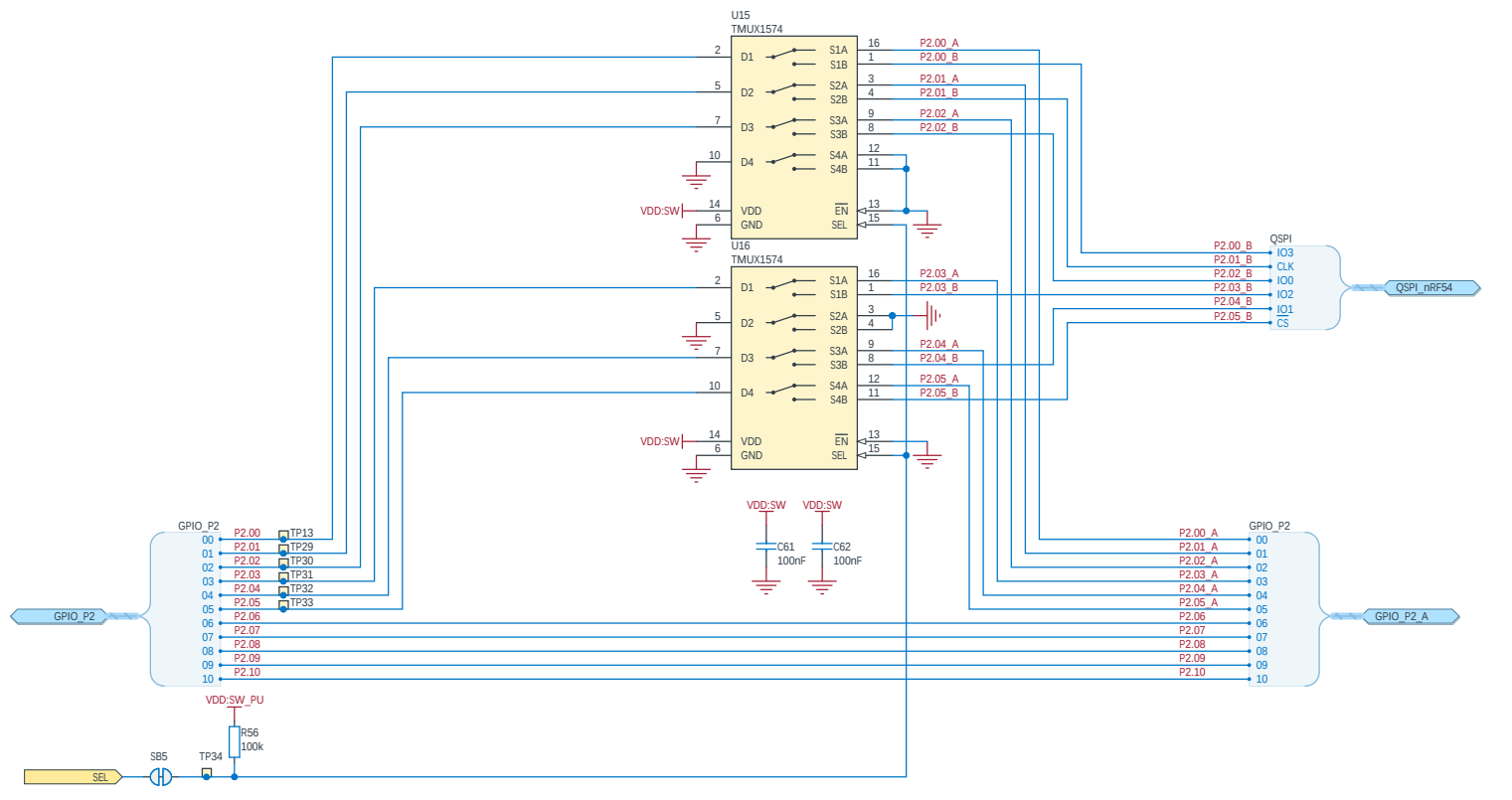
The VDD:nRF buffer is not needed for nRF54LM20 reference design. Only used for DK operation and peripherals



Title nRF54LM20 Development Kit - Power Supply		
Size A3	PCB Assembly Number PCA10184	Revision 0.7.0
Date: 10.02.2026	File: pca10184_power_supply.SchDoc	Sheet 6 of 9
Classification: PUBLIC	Drawn By: GETR	

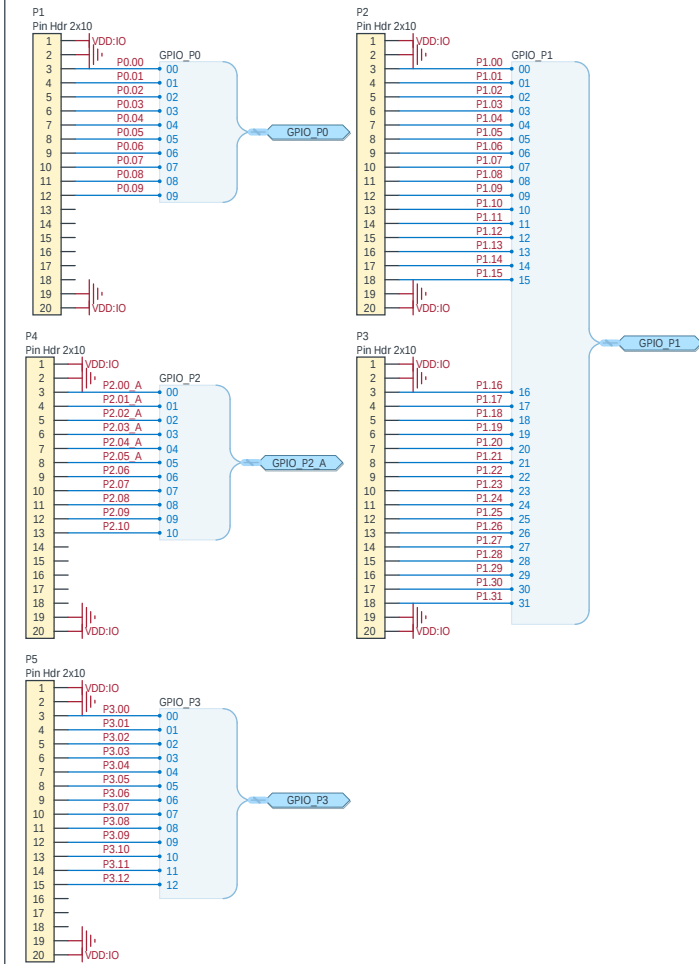
QSPI switching, GPIO Port 2

[Datasheet link \(TMUX1574\)](#)

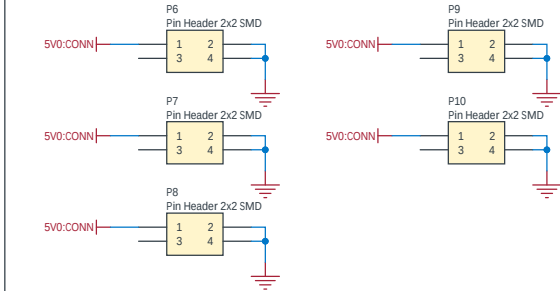


Title nRF54LM20-DK - QSPI Switching			
Size A3	PCB Assembly Number PCA10184	Revision 0.7.0	
Date: 10.02.2026		Sheet 7 of 9	
File: pca10184_switches.SchDoc		Drawn By: GETR	
Classification: PUBLIC			

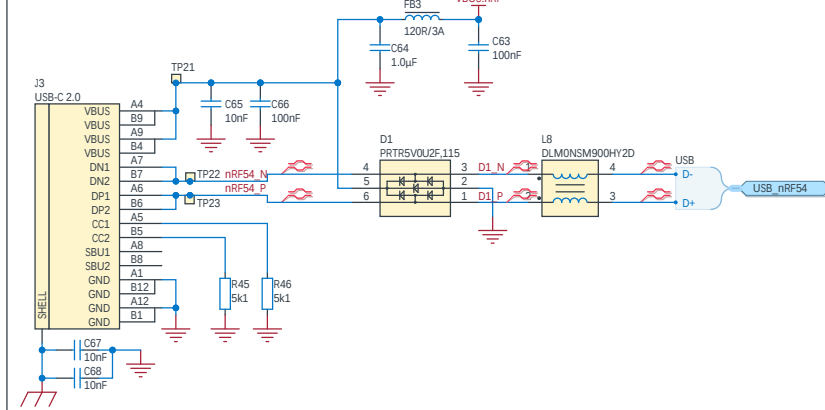
GPIO Connectors



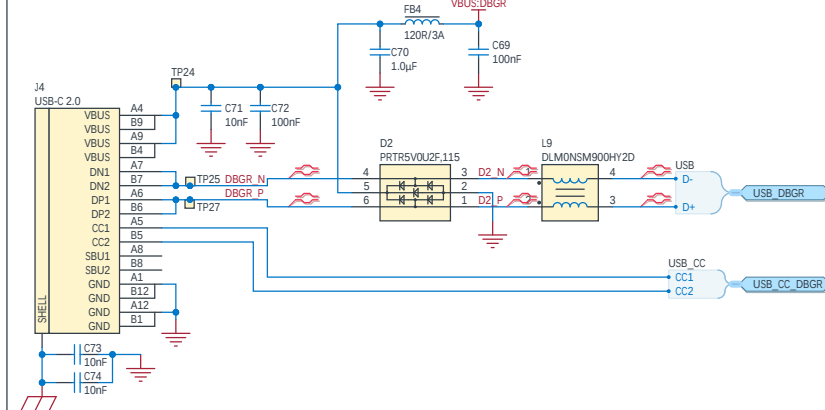
Power Connectors



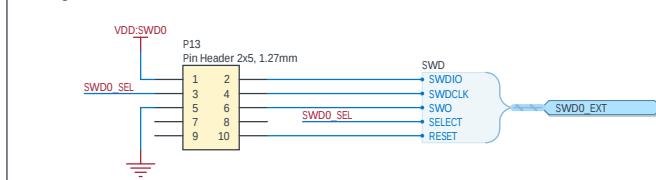
nRF USB Connector



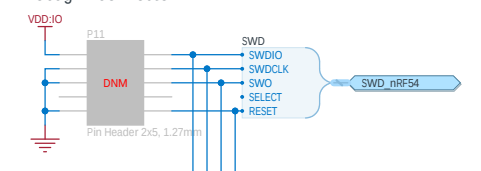
Debugger USB Connector



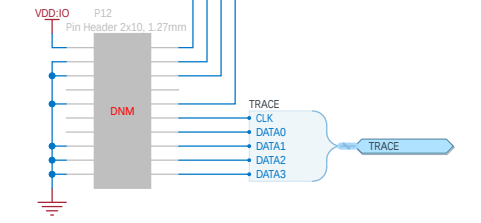
Debug OUT Connector



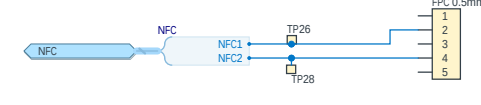
Debug IN Connector



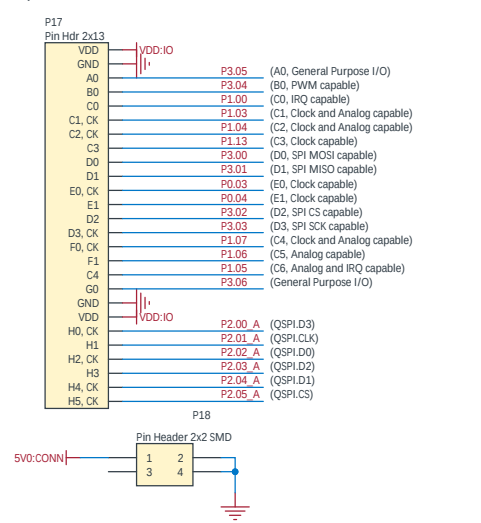
Trace connector



NFCT Antenna Connector



Expansion Board Header



Title
nRF54LM20 Development Kit - Connectors

Size
A3

PCB Assembly Number
PCA10184

Date: 10.02.2026
File: pca10184_connectors_SchDoc
Classification: PUBLIC

Revision
0.7.0

Sheet 8 of 9
Drawn By: GETR



The No ERC object is a design directive.
This directive is placed on a node in the circuit to suppress harmless warnings and/or error violation conditions that are detected when the schematic project is compiled.

Board fiducials



Logos and markings

PCA10184
0.7.0
<year>. <week>
<UID>


WEEE Symbol


It simply works!

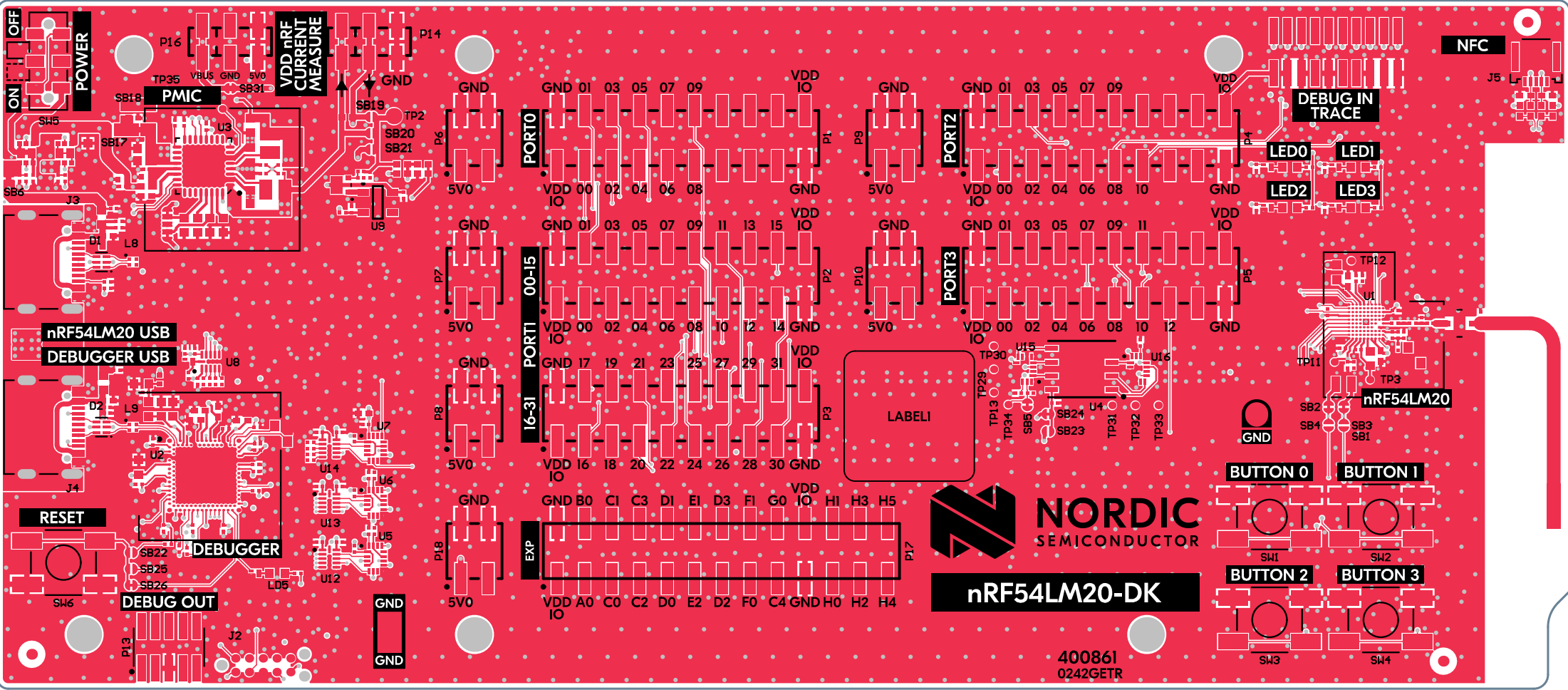
Mechanical holes

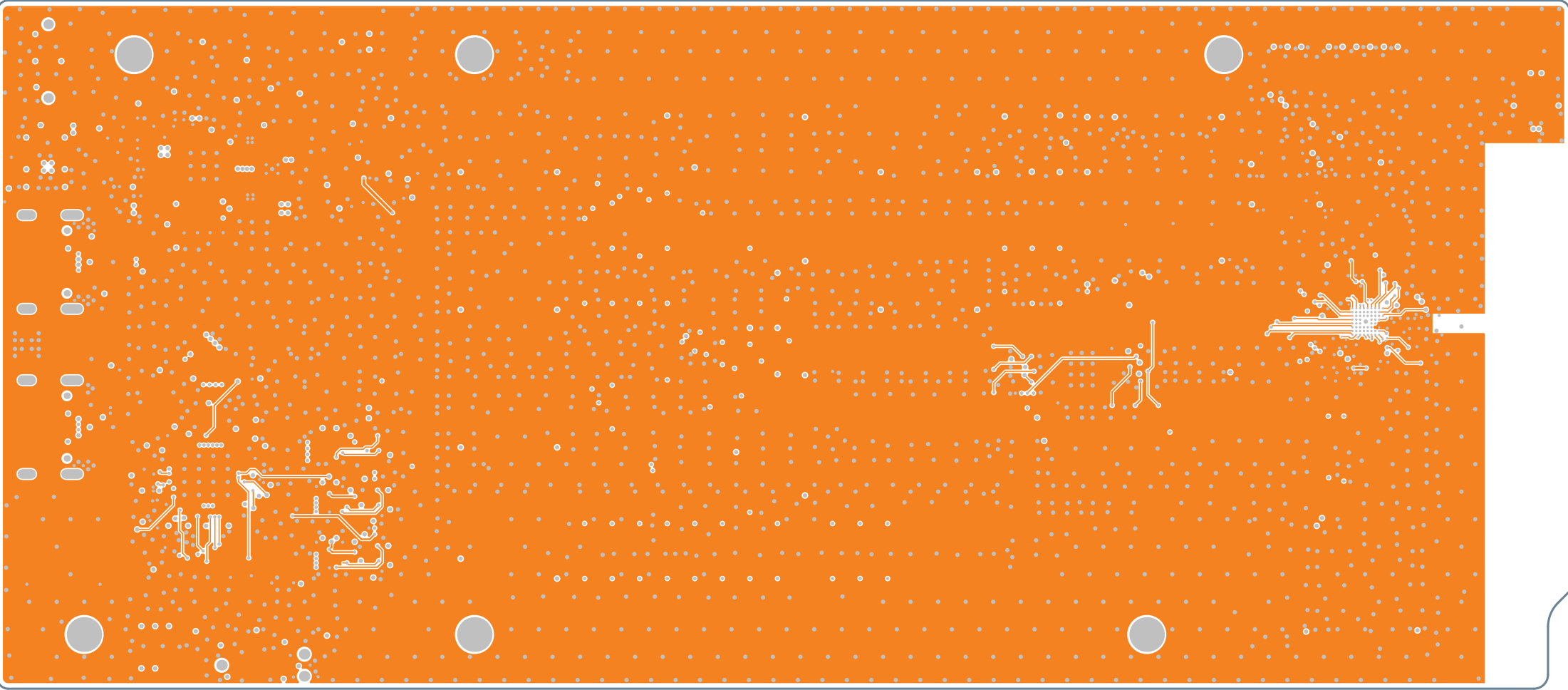


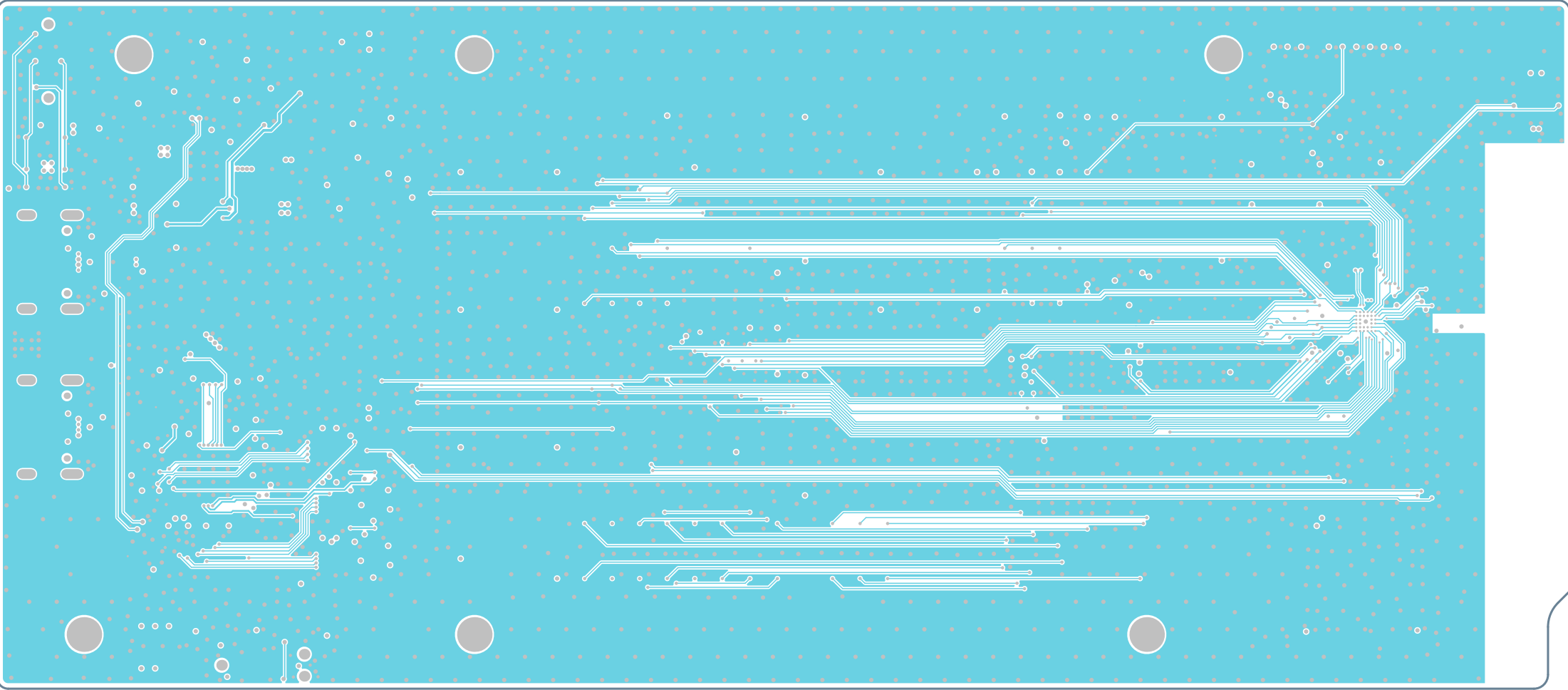
H1 H2 H3
H4 H5 H6

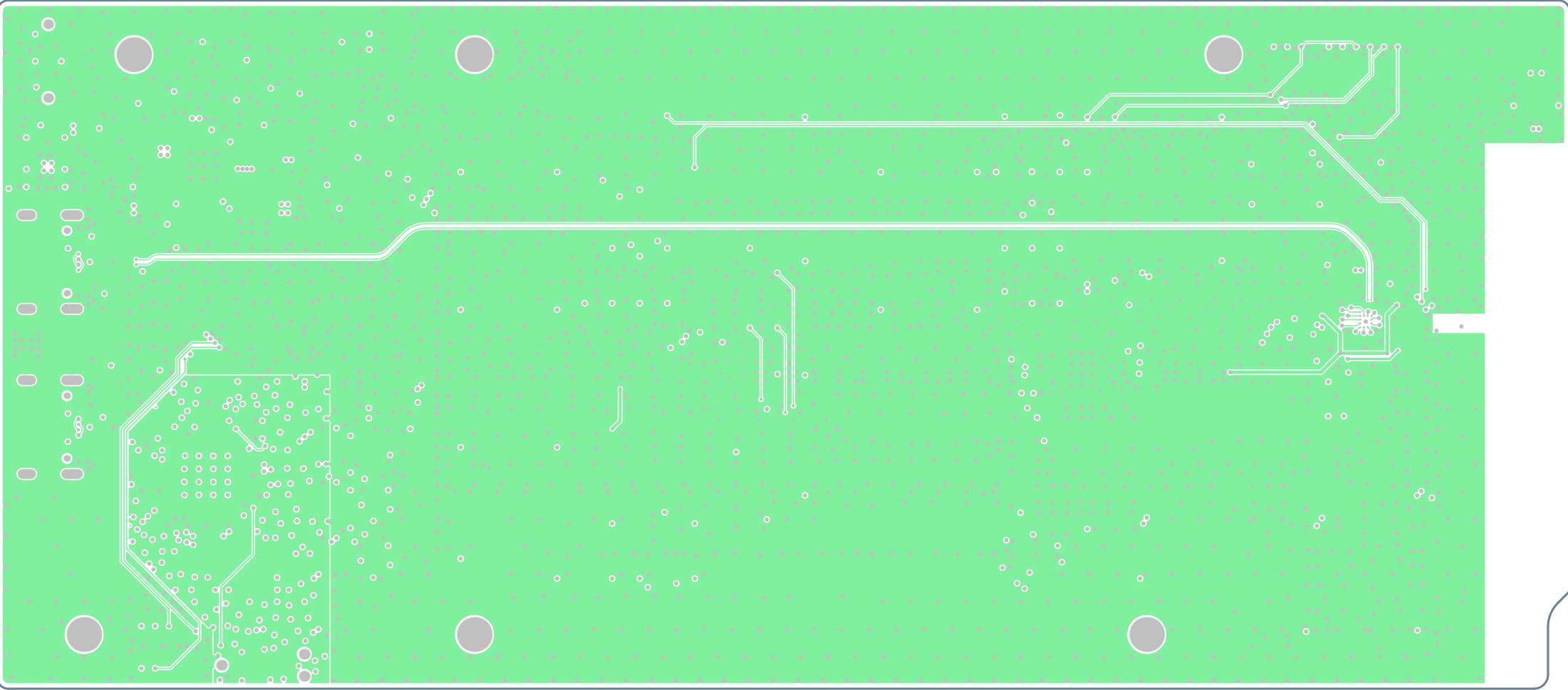
Title nRF54LM20 Development Kit - Mechanical		
Size A4	PCB Assembly Number PCA10184	Revision 0.7.0
Date: 06.03.2026		Sheet 9 of 9
File: pca10184_mechanical.SchDoc		Drawn By: GETR
Classification: PUBLIC		

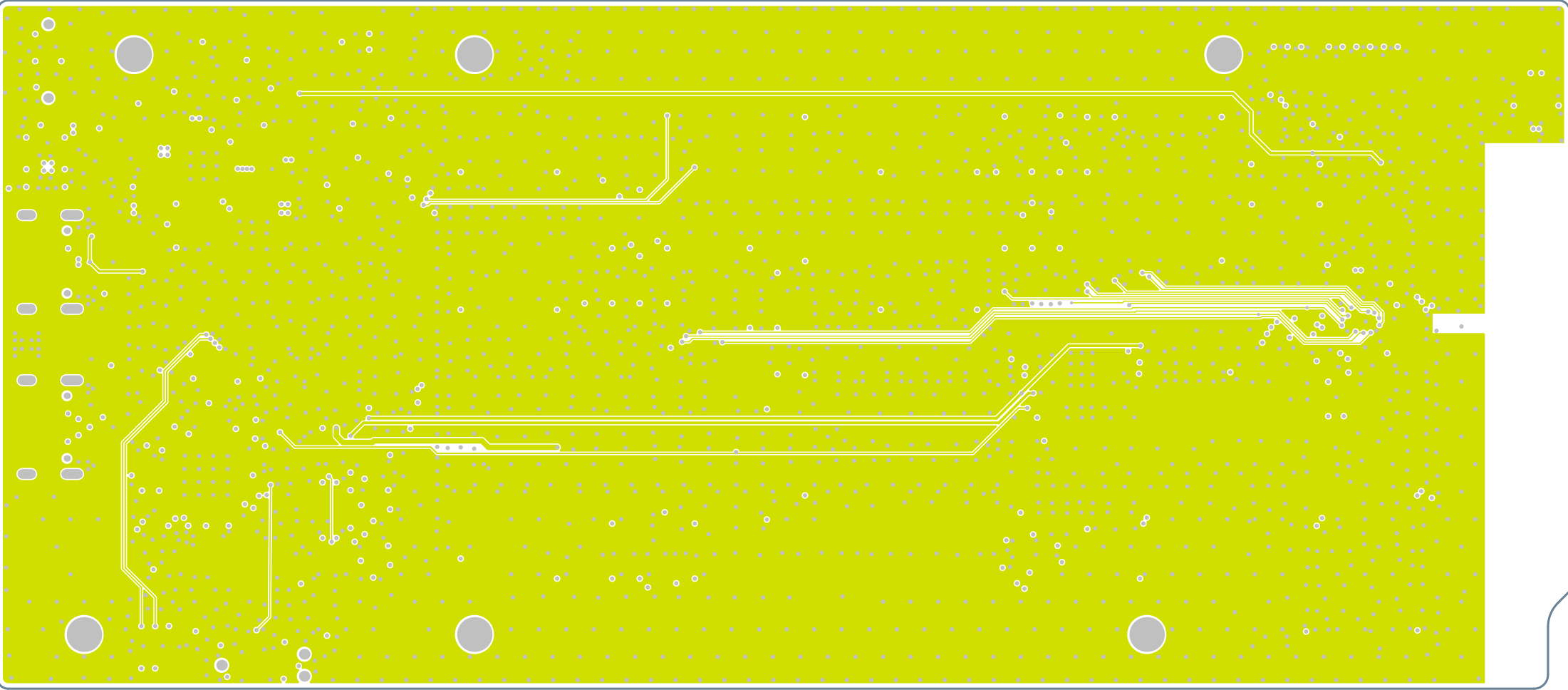


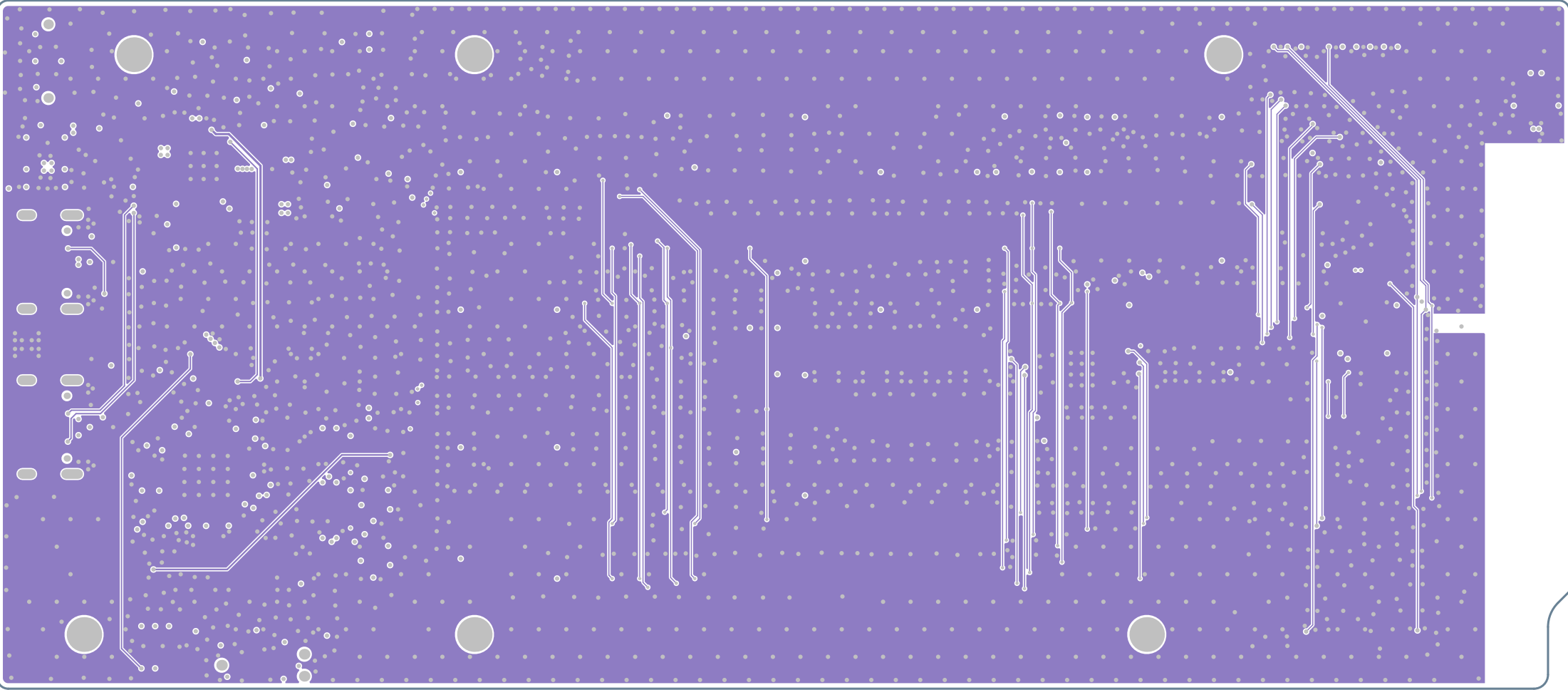


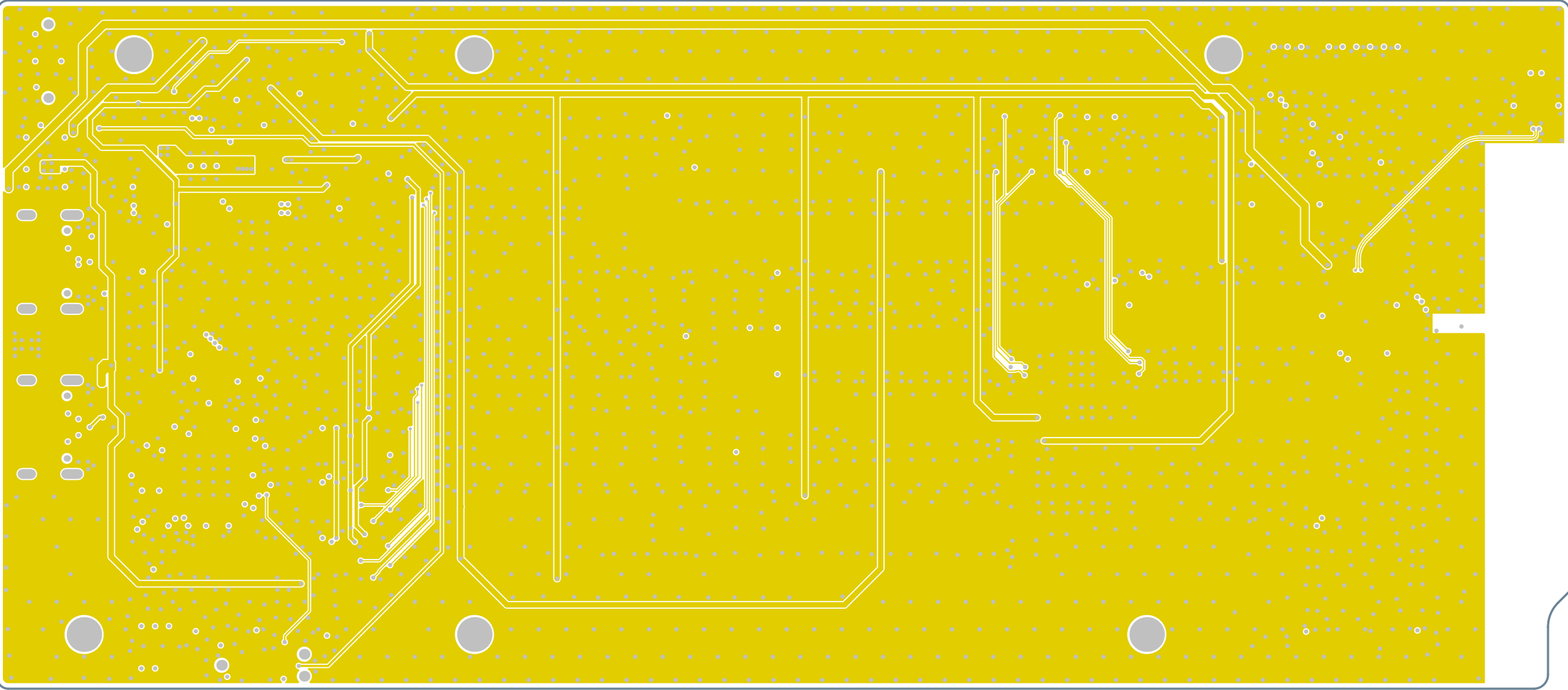




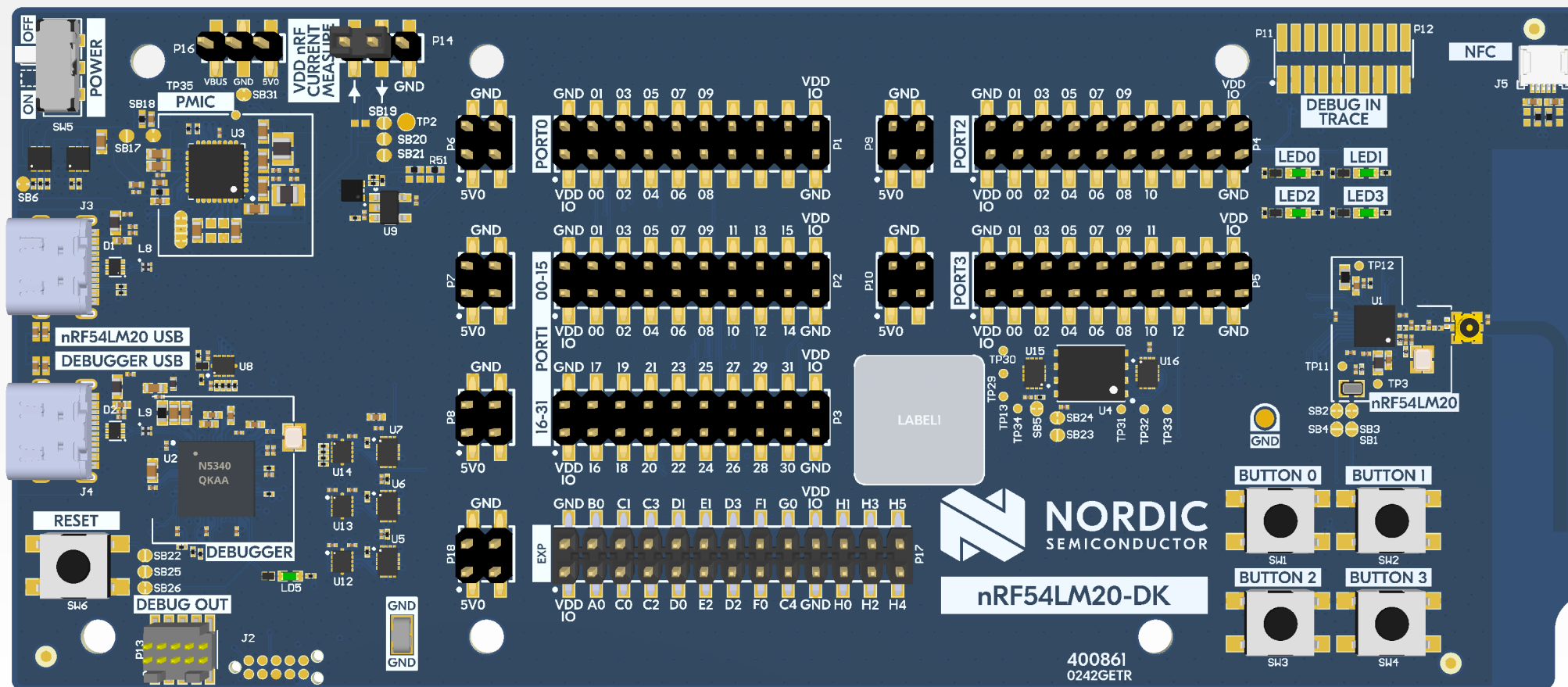














A circuit diagram showing a 5.0V DC voltage source connected in series with a diode and a resistor labeled W1. The diode is oriented with its cathode towards the positive terminal of the voltage source. The resistor W1 is connected in series with the diode. The circuit is shown on a blue background.

